

CS333 Midterm 2 Part (a)

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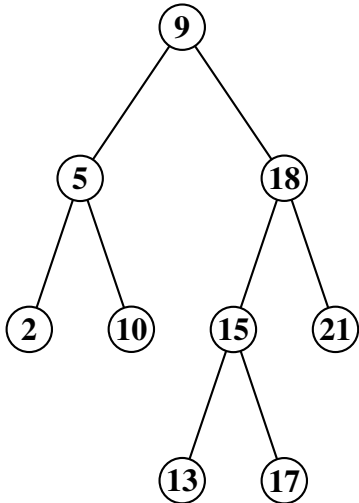
I. MULTIPLE CHOICE QUESTIONS (40 POINTS)

1. If we want to obtain a rightly imbalanced binary search tree from numbers 1 to $2^n - 1$, what is the correct order of insertion of those numbers to the binary search tree?

- a) $2^n - 1, \dots, 2, 1$
- b) $1, 2, \dots, 2^n - 1$
- c) $2^{n-1}, 2^{n-2}, \dots, 1$
- d) $1, 3, 5, \dots, 2, 4, 6, \dots$
- e) $2^{n-1}, 2^{n-2}, 2^{n-1} + 2^{n-2}, \dots, 2^n - 1$

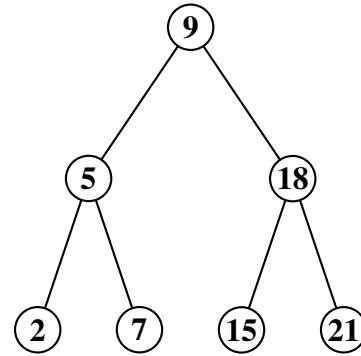
2. What is the best, worst time complexity for insertion in a fully balanced binary search tree?

- a) $\mathcal{O}(1), \mathcal{O}(\log N)$
- b) $\mathcal{O}(1), \mathcal{O}(N)$
- c) $\mathcal{O}(1), \mathcal{O}(1)$
- d) $\mathcal{O}(\log N), \mathcal{O}(\log N)$
- e) $\mathcal{O}(\log N), \mathcal{O}(N)$



3. Given the tree above, which node does not satisfy the binary search tree property?

- a) 18
- b) 10
- c) 13
- d) 21
- e) 17



4. For the tree given above, where can we insert 14?

- a) Right child of 7
- b) Left child of 7
- c) Right child of 15
- d) Left child of 15
- e) Left child of 21

5. What is the average and worst case time complexity of Merge sort?

- a) $\mathcal{O}(N), \mathcal{O}(N)$
- b) $\mathcal{O}(N), \mathcal{O}(N \log N)$
- c) $\mathcal{O}(N \log N), \mathcal{O}(N \log N)$
- d) $\mathcal{O}(N \log N), \mathcal{O}(N^2)$
- e) $\mathcal{O}(N^2), \mathcal{O}(N^2)$

6. What is the average and worst case time complexity of Quick sort?

- a) $\mathcal{O}(N), \mathcal{O}(N)$
- b) $\mathcal{O}(N), \mathcal{O}(N \log N)$
- c) $\mathcal{O}(N \log N), \mathcal{O}(N \log N)$
- d) $\mathcal{O}(N \log N), \mathcal{O}(N^2)$
- e) $\mathcal{O}(N^2), \mathcal{O}(N^2)$

7. Algorithm C solves problems of size N by dividing them into nine subproblems of size $N/3$, recursively solving each subproblem, and then combining the solutions in $\mathcal{O}(N^2)$ time. What is the time complexity of C?

- a) $\mathcal{O}(N^2 \log N)$
- b) $\mathcal{O}(N^2)$
- c) $\mathcal{O}(N^3)$
- d) $\mathcal{O}(N \log N)$
- e) $\mathcal{O}(N)$

8. Algorithm B solves problems of size N by recursively solving four subproblems of size $N/2$ and then combining the solutions in constant time. What is the time complexity of B?

- a) $\mathcal{O}(N^2 \log N)$
- b) $\mathcal{O}(N^2)$
- c) $\mathcal{O}(N^3)$
- d) $\mathcal{O}(N \log N)$
- e) $\mathcal{O}(N)$

9. In the multiplication of two n by n matrices, the standard divide and conquer approach divides matrices into 4 equal parts, and then it multiplies and adds these submatrices to get the end result. The number of multiplications required to get the result matrix is 8 and the internal matrices are added in $\mathcal{O}(N^2)$ time. Let say you found out a new algorithm with 6 multiplications instead of 8, then your new algorithm will have the time complexity of?

- a) $\mathcal{O}(N^3)$
- b) $\mathcal{O}(N \log N)$
- c) $\mathcal{O}(N^{\log_4 6})$
- d) $\mathcal{O}(N^{\log_2 6})$
- e) $\mathcal{O}(N^2)$

10. In the multiplication of two n by n bits integers, the standard divide and conquer approach divides integers into 2 equal parts, and then it multiplies and adds these binary integers to get the end result. The number of multiplications required to get the result integer is 4 and the internal numbers are added in $\mathcal{O}(N)$ time. Let say you found out a new algorithm with 2 multiplications instead of 4, then your new algorithm will have the time complexity of?

- a) $\mathcal{O}(N^2)$
- b) $\mathcal{O}(N \log N)$
- c) $\mathcal{O}(N^{\log_4 2})$
- d) $\mathcal{O}(N^{\log_2 6})$
- e) $\mathcal{O}(N)$

11. Let say we want to sort the following integers 2, 12, 6, 4, 15, 7, 1, 10. Let say the pivot is selected as 10, which of the following shows a possible list of these numbers after the pivot is put into its correct place?

- a) 10, 12, 6, 4, 15, 7, 1, 2
- b) 1, 2, 4, 5, 6, 7, 12, 15, 10
- c) 2, 6, 4, 7, 1, 10, 15, 12
- d) 2, 12, 6, 10, 4, 15, 7, 1
- e) 2, 12, 6, 4, 15, 7, 1, 10

12. What is the edit distance between words 'ankara' and 'akraba'?

- a) 2
- b) 3
- c) 4
- d) 5
- e) 6

13. What are the values of the dynamic programming table for the cells 3, 4, 5, 6 for the Coin Change problem with the denominations 1, 3, 4?

- a) 1, 1, 1, 2
- b) 1, 1, 1, 1
- c) 1, 2, 1, 2
- d) 2, 1, 2, 2
- e) 1, 1, 2, 2

14. How many paths are there from cell (1, 1) to cell (4, 5) in a rectangular matrix, if the only possible moves are going right and going down?

- a) 15
- b) 20
- c) 35
- d) 45
- e) 70

15. How many paths are there from cell (1, 1) to cell (4, 5) in a rectangular matrix, if the only possible moves are going right, going down, going diagonal?

- a) 41
- b) 63
- c) 81
- d) 129
- e) 154

16. How many function calls are needed for recursive calculation of Fibonacci(5), if Fibonacci(1) and Fibonacci(0) are known?

- a) 8
- b) 12
- c) 13
- d) 14
- e) 21

17. What is the size of the longest increasing subsequence for 10, 22, 9, 33, 21, 50, 41, 60, 80?

- a) 3
- b) 4
- c) 5
- d) 6
- e) 7

18. What is the largest contiguous subarray sum for -2, -3, 4, -1, -2, 1, 5, -3?

- a) 3
- b) 4
- c) 5
- d) 6
- e) 7

19. Given the matrices A, B, C, D with sizes 1×2 , 2×10 , 10×3 and 3×5 , which of the following order of multiplication requires minimum multiplications?

- a) $A \times ((B \times C) \times D)$
- b) $(A \times B) \times (C \times D)$
- c) $((A \times B) \times C) \times D$
- d) $(A \times (B \times C)) \times D$
- e) $A \times (B \times (C \times D))$

20. Given the weights and values in knapsack problem, what is the maximum total value for a knapsack of weight 11?

i	v_i	w_i
1	1	1
2	6	2
3	18	5
4	22	6
5	28	7

- a) 30
- b) 35
- c) 40
- d) 45
- e) 50